

Review

Comparative Analysis of Electrochemical and Thermochemical Hydrogenation of Biomass-Derived Phenolics for Sustainable Biofuel and Chemical Production

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Abstract: The electrocatalytic hydrogenation (ECH) of biomass-derived phenolic compounds is a promising approach to the production of value-added chemicals and biofuels in a sustainable way under moderate reaction conditions. This study provides a comprehensive comparison of electrochemical and thermochemical hydrogenation processes, highlighting their relative advantages in terms of energy efficiency, product selectivity, and environmental impact. Several electrocatalysts (Pt, Pd, Rh, Ru), membranes (Nafion, Fumasep, GO-based PEMs), and reactor configurations are tested for the selective conversion of model compounds such as phenol, guaiacol, furfural, and levulinic acid. The contributions made by the electrode material, electrolyte composition, membrane nature, and reaction conditions are critically evaluated in relation to Faradaic efficiency, conversion rates, and product selectivity. The enhancement in the performance achieved by a new catalyst architecture is emphasized, such as MOF-based systems and bimetallic/trimetallic catalysts. In addition, a demonstration of graphite-based membranes and membrane-separated slurry reactors (SSERs) is provided, for enhanced ion transport and reaction control. The results illustrate the potential of using ECH as a low-carbon, scalable, and tunable method for the upgrading of biomass. This study offers valuable insights and guidelines for the rational design of next-generation electrocatalytic systems toward green chemical synthesis and emphasizes promising perspectives for the strategic development of electrochemical technologies in the pathway of a sustainable energy economy.

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1. Introduction

In recent years, considerable studies have clearly proved the possibility of utilizing biomass fractions as renewable sovereign materials for the liquid fuels, organic chemicals, and raw materials for various industries. These studies report the dependence of various organics in the design of sustainable sources of energy and materials and how they diminish the need for fossil fuels in the development of a decarbonized world [1]. The controlled and selective transformation of biomass-derived components into specific organic molecules and synthetic organic polymers with carbon–carbon bonds underscores the importance of sustainable chemical production processes. These changes offer not only economic benefits but also a reduced environmental impact. These processes increase the efficient use of feedstocks and promote the development of renewable-based production, which is a first step for the development of sustainable replacements of petroleum-based chemicals [2].

Chemical-catalytic processes are a high-value-added technology for the processing of biomass products.

These approaches possess fast reaction times that range from a few seconds to hours and permit selective conversions into specific products. In addition, they are becoming more and more attractive because of their scaling, raw material utilization, low-cost, and energy-efficient processes. Given the compatibility of these technologies with the current infrastructure of oil refineries, there is a substantial potential to speed up the transformation of the chemical industry to the production of chemicals from biomass and to create a sustainable chemical industry [3]. In the emerging scenario, specially designed chemical technologies are being actively developed for the efficient and well-controlled conversion of chemical components of biomass into organic chemicals, having a predetermined structure and/or functionality. These initiatives aim to produce high-value-added chemical products originating from renewable raw materials and serving as sustainable alternatives to fossil-based chemicals [4].

In comparison to fossil-based resources, biomass can be grown in diverse climatic and geographic regions, leading to a large raw material potential with particular properties that depend on the location. Furthermore, a major fraction of urban and industrial waste streams are biomass-based [5,6]. Biomass as a source of both energy and non-food organic chemicals, including bioactive molecules, has historically played a pivotal role long predating the advent of petroleum. However, the consolidated bioprocessing of biomass to generate the full suite of major organic chemical classes and value-added products through a biorefinery concept remains largely underdeveloped as a field of study. The high oxygen content of bio-based materials, structurally different from petroleum-produced hydrocarbons, poses both technical challenges and opportunities in obtaining biorefinery products [7,8].

Biomass largely consists of an interconnected network of polymeric compounds, most of which contain functional groups and are highly oxygenated. Selective modification to recover the target compounds with certain functional groups is an important issue. In order to overcome this limitation, the application of fractionation techniques, which break complex structures down into their components, in combination with chemical means of re-assembling or functionalizing target molecules is gaining importance [9]. Several molecules derived by biomass processing have high functionality and carbon atoms in different oxidation states. As a result, a major challenge is the site-selective functionalization of these functional groups in order to tune the carbon backbone for specific reactions. Moreover, as the majority of biomass components are polymeric, a previous depolymerization or separation stage frequently needs to be performed before chemical conversion. Improving biorefinery efficiency through the adaptation of petroleum-refining processes and the use of available industrial databases is reasonable and promising, being encouraged by Patel et al. [10]. Carbocyclic compounds are widely used in industry as basic materials for the production of various kinds of commercial organic compounds [11]. The chemocatalytic conversion of biomass to carbocyclic compounds broadens the range of renewable organic compounds and greatly improves the economics of biorefinery processes [12]. At present, the majority of accessible carbocyclic materials have their origin in petroleum feedstocks. For instance, oxidation is commonly used in petrochemical refineries to insert functional groups into hydrocarbons. However, since platform molecules from biomass inherently contain oxygen and reactive functional groups, many of these oxidations steps in biomass-based processes may be unnecessary. Yet there are still many technical barriers to overcome before such sustainable and atom-economic access to carbocyclic skeletons from biomass is possible.

Biomass (plant- and microbial-derived) is a renewable and biodegradable source of carbon. Its content depends on its origin, but its major constituents are structural

biopolymers including cellulose, hemicellulose, and lignin. Cellulose is the most abundant natural plant biomass, consisting of a linear polysaccharide chain that contains D-glucose monomers linked by β -1,4-glycosidic bonds and constituting a major component of the plant cell wall and providing its structural framework. Its crystalline structure, stabilized by extensive hydrogen bonding, provides significant rigidity and regularity. Cellulose is composed of crystalline regions (which give cellulose its strength) and amorphous regions (which gain reactivity). However, cellulose is resistant to water, organic solvents but can be degraded with strong acids or bases and hydrolyzed to glucose by cellulase enzymes [13]. Hemicellulose, which is the other constituent of plant cell walls, is more heterogeneous and branched, formed of pentoses (xylose or arabinose, for example) and hexoses (mannose or galactose). It has a lower molecular weight and crystallinity than cellulose and is more sensitive to being hydrolyzed; meanwhile, the amorphous property of this material is likely to be achieved for chemical degradation and conversion to fermentable sugars under acidic or alkaline conditions [14]. It is the lignin which, enveloping the cellulose and hemicellulose, confers plant cell walls with structural resistance and hydrophobic features. It is made of phenylpropanoid units (such as p-hydroxyphenyl, guaiacyl, and syringyl groups) and is highly thermal stable, and can be crosslinked with heating [15]. Biomass also contains extractive substances such as lipids, proteins, volatile organic compounds, and inorganics. Bacteria have a high content of protein of value for enzyme production in biomass. Lipid classes such as triglycerides (TAGs), free fatty acids (FFAs), and phospholipids (PLs) are significant components required for biodiesel production from algal biomass. Phenolics, essential oil, and terpenoids are some substances mostly used in the drug and cosmetic industries. Ash content and mineral nutrients, which include carbon, calcium, potassium, and magnesium, are elements that differ between plant species [2].

2. Thermochemical Conversion Processes for Biomass Valorization

The thermochemical conversion process, together with its associated cycle, forms a complex and multifaceted pathway that is critical in today's energy landscape. As illustrated in Figure 1, the production route for bio-oil, often referred to as bio-crude oil, and advanced value-added biofuels sourced from biomass are positioned within the broader framework of a circular biorefinery approach. This innovative process is not just about producing energy; at its core, it is fundamentally reliant on the generation of sustainable energy derived from renewable resources, which plays a vital role in reducing carbon footprints and contributing to a more sustainable future. This approach leverages biomass to encourage the effective utilization of resources while highlighting the significance of circularity within the economy.

The cycle commences with the conversion of solar energy into plant biomass through the process of photosynthesis. In this natural process, trees and other vegetation harness sunlight to transform carbon dioxide (CO_2) and water into structural biopolymers, namely cellulose, hemicellulose, and lignin. The harvested biomass undergoes appropriate processing and is subsequently fractionated into biochemical components that serve as feedstock for thermochemical methods. Employing techniques such as hydrothermal liquefaction (HTL), pyrolysis, or gasification, the biomass is subjected to elevated temperatures and pressures to yield bio-oil and other platform chemicals.

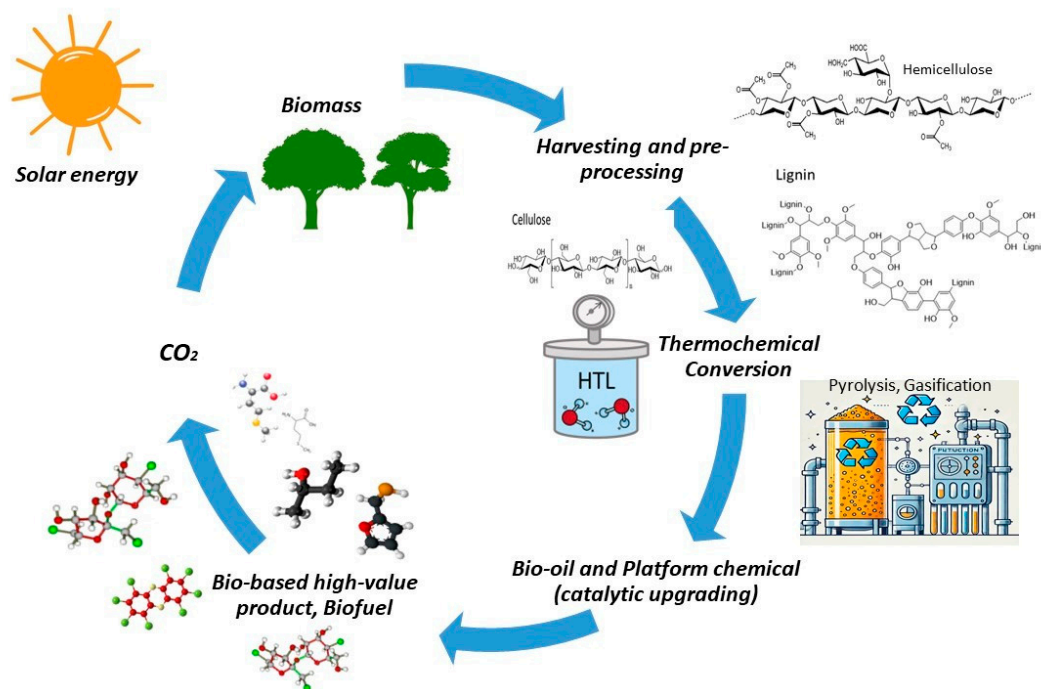


Figure 1. Biomass-to-biofuel: a thermochemical conversion and catalytic upgrading cycle.

These thermochemical methods disrupt the molecular architecture of the biomass, resulting in energy-dense liquid products. The bio-oil produced is then subjected to catalytic upgrading processes, transforming it into advanced biofuels (e.g., biodiesel, green gasoline) and a variety of high-value chemicals. This step facilitates the generation of high-quality substitutes for conventional fossil fuels. Upon utilization, the released CO₂ is reabsorbed from the atmosphere via photosynthesis, effectively completing a carbon-neutral cycle. In contrast to fossil fuels, this closed-loop system significantly mitigates environmental impacts and bolsters the development of sustainable energy infrastructures. The catalytic depolymerization of carbohydrates enables the synthesis of diverse heterocyclic and carbocyclic compounds. For example, the acid-catalyzed pyrolysis of cellulose results in the formation of levoglucosenone (LGO), whereas acid hydrolysis produces 5-(hydroxymethyl)furfural (HMF) [15]. Likewise, pentose sugars derived from hemicellulose undergo hydrolysis and dehydration under acidic conditions to yield furfural (FUR). Both HMF and FUR can be generated directly through the catalytic degradation of polymeric carbohydrates or via preliminary catalytic depolymerization followed by the conversion of isolated simple sugars [16]. In this context, synthesizing carbocyclic compounds from renewable carbon sources has become a significant area of research. Established transformations in synthetic organic chemistry facilitate the design of carbocyclic molecules with desirable structural and functional attributes. Reactions such as aldol and Knoevenagel condensations are utilized to elongate carbon backbones and promote the formation of carbocyclic core structures. Furthermore, carbon–carbon bond-forming reactions, including the Diels–Alder reaction, are essential for constructing cyclic frameworks [17,18].

Hydrothermal Liquefaction (HTL) of Biomass: Principles, Advantages, and Challenges.

Hydrothermal liquefaction (HTL) is a thermochemical conversion technique that converts biomass or organic waste into liquid bio-crude oil through the application of high temperature and pressure in the presence of water. The efficacy of HTL depends on operational parameters such as temperature, pressure, reaction time, and feedstock characteristics. Typically, HTL operates at temperatures between 250 and 400 °C and pressures of 10–25 MPa [19]. Lower temperatures (250–300 °C) favor oxygen-rich compounds,

while higher temperatures (350–400 °C) yield lower-oxygen, more energy-dense bio-oils. Under high temperature and pressure, biomass structures such as cellulose and lignin break down more easily, thus raising the yield of liquid. But temperatures that are too high may also bring on undesirable gases along with solid byproducts, which together decrease overall fuel output [20]. Time of reaction directly determines the yield and quality of bio-oil. Shorter times typically favor liquid yields, while long durations often lead to greater byproducts. Optimal reaction times usually fall in the range from 15 to 60 min, because prolonged exposure may cause oxidative decay. High moisture content is advantageous, as water enhances reaction kinetics and conversion rates [21]. During HTL, water acts as a solvent under subcritical and supercritical conditions, facilitating key reactions like hydrolysis and dehydration that promote bio-oil formation. The resulting bio-oils can serve as renewable alternatives to fossil fuels in established systems, contributing to carbon neutrality and reducing greenhouse gas emissions.

These bio-oils consist of a complex mixture of organic compounds, including phenols and acids, which hold value for industries such as plastics and pharmaceuticals [22]. Despite its potential, HTL faces technical and economic challenges. High temperatures and pressures can lead to significant energy consumption, impacting feasibility. Also, specialized reactor systems are required to withstand extreme conditions, adding to costs. Catalyst performance also poses challenges; issues with deactivation or loss increase operational expenses, stressing the importance of catalyst recyclability and stability. Furthermore, the bio-oil's high oxygen content and acidity often necessitate additional upgrading processes. The production of gas and solid byproducts can hinder process efficiency and complicate waste management. Safety concerns associated with high-pressure operations necessitate robust design and maintenance strategies. In conclusion, hydrothermal liquefaction presents a viable method for converting wet and low-value biomass into energy.

Pyrolysis of Biomass and Organic Waste: Process Overview, Products, and Challenges.

Pyrolysis is a thermochemical process that converts biomass and organic waste into biochar, bio-oil, and syngas through controlled thermal decomposition at high temperatures in an oxygen-free environment. It plays a crucial role in waste management, renewable energy production, and carbon sequestration. Pyrolysis enhances the valorization of various organic waste types while mitigating environmental impacts. The process begins with the evaporation of moisture from the feedstock at temperatures of 100 to 200 °C, which is essential for improving the efficiency of subsequent thermochemical reactions. The main pyrolysis occurs at temperatures between 400 and 800 °C, where organic materials break down into simpler compounds [23]. The products of pyrolysis include biochar, a carbon-rich solid residue; bio-oil, an energy-dense liquid; and syngas, a mixture of carbon monoxide (CO), hydrogen (H₂), methane (CH₄), and other light hydrocarbons [21]. Biochar can be further activated using carbon dioxide (CO₂) or steam to enhance its adsorption capacity and surface area, making it suitable for applications like water purification and soil amendment.

Bio-oil can be refined into biodiesel and bio-jet fuel, serving as a source for chemicals like phenols. Syngas can generate electricity or produce synthetic fuels via Fischer–Tropsch synthesis. Pyrolysis provides a carbon-neutral alternative to fossil fuels, helping to reduce greenhouse gas emissions and recover organic wastes, including plastics and agricultural residues, thus limiting environmental pollution. It accommodates various biomass and waste types, such as lignocellulosic biomass and plastics. However, pyrolysis faces challenges, including high energy consumption due to necessary temperatures and rapid heating rates. Special reactors and high-performance materials raise capital costs and economic viability issues. Raw bio-oil's high oxygen content and acidity necessitate hydrotreatment and refinement for fuel use. The quality of biochar varies with feedstock

and process parameters, and separating byproducts like syngas and tars incurs additional costs [24].

Gasification of Biomass and Organic Waste: A Sustainable Route for Syngas and Hydrogen Production.

Gasification is a thermochemical process that converts carbonaceous materials such as biomass, coal, and organic waste into synthesis gas (syngas, which is composed mainly of hydrogen (H₂) and carbon monoxide (CO)). This conversion takes place at very high temperatures with controlled amounts of oxygen or steam. Gasifiers have been utilized in applications such as waste-to-energy, power generation, and synthetic fuel production, as a cleaner and more efficient alternative to combustion-based processes. Its performance depends on the process conditions of temperature, pressure, and oxygen/steam ratio.

Typical gasification temperatures range from 700 to 1300 °C. Lower temperatures (700–900 °C) tend to produce more tars and heavy hydrocarbons and exhibit slower reaction kinetics, resulting in lower syngas yields. Conversely, medium temperatures (900–1100 °C) are often optimal, striking a balance between high syngas production and minimal tar formation. At higher temperatures (1100–1300 °C), tars are fully converted into the gas phase, yielding cleaner syngas, though this also raises energy demands and the risk of corrosion in reactor materials [21,25]. Operating pressures are generally atmospheric or slightly elevated (20–40 bar); higher pressures can increase syngas energy density but complicate reactor design and raise costs. The careful control of oxygen and steam input is critical. While oxygen facilitates partial oxidation, an excess can lead to complete combustion and reduced efficiency through increased CO₂ production. Steam enhances hydrogen content and suppresses tar formation, but an excess can also increase energy consumption [26,27].

Molecular rearrangements are employed to modify existing carbon skeletons, while hydrodeoxygenation (HDO) reactions diminish the oxygen content, resulting in the production of more hydrophobic and stable compounds. Biomass-derived carbocyclic compounds may, in certain instances, share identical structures with their petroleum-derived counterparts, while in other cases, they may serve as functional equivalents within the chemical industry. In both scenarios, these compounds can be chemically modified and optimized for specific industrial applications. However, it is crucial to recognize that carbocyclic compounds cannot be synthesized directly from sugars or polymeric carbohydrates via enzymatic pathways. Consequently, their production from biomass relies solely on thermochemical or chemocatalytic conversion processes.

The formation of carbocyclic compounds is influenced by the molecular structure of the carbohydrates used. For example, pentose sugars (e.g., xylose) tend to produce cyclopentane-based compounds, while hexoses may yield either cyclopentane or cyclohexane derivatives. The synthesis of larger carbocyclic compounds typically involves the interaction of multiple intermediate chemical species through a series of controlled chemical transformations. Although significant attention has recently been attributed to the catalytic synthesis of various carbocyclic compounds derived from biomass, the existing body of literature remains fragmented and lacks a holistic overview. This study aims to integrate current knowledge, evaluate recent advancements, and address critical challenges in the synthesis of carbocyclic compounds from biomass. Through this effort, the study seeks to provide researchers with enhanced insights into sustainable chemical production strategies rooted in biomass while identifying potential avenues for future investigation.

Figure 2 illustrates the various pathways through which lignocellulosic biomass can be converted into energy and value-added products. These pathways are categorized into two primary types: thermochemical and biochemical methods. Thermochemical methods involve treating biomass at high temperatures, often following initial processes

such as drying, fast or slow pyrolysis, pelletization, and torrefaction. This can lead to liquefaction and gasification, which produce fuel and gas products; pyrolysis, which generates fuels, gases, and chemical substances; and combustion, which creates heat and electricity. On the other hand, biochemical methods utilize microorganisms or enzymes for biomass conversion. Following pretreatment and hydrolysis, fermentation processes result in bioethanol, biobutanol, and various chemicals, while esterification produces biodiesel and anaerobic digestion generates biogas. Figure 2 provides a comprehensive summary of these conversion pathways and their resulting products, highlighting the variety and potential for sustainability within biomass conversion techniques.

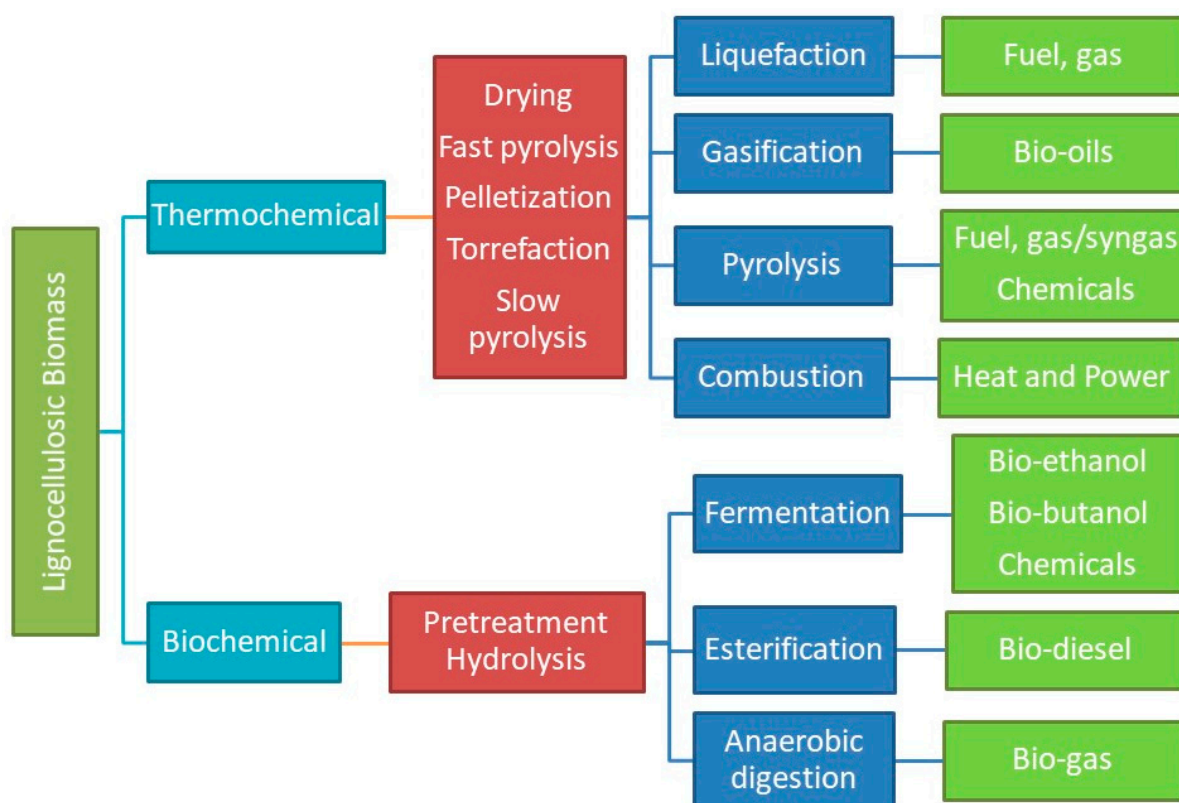


Figure 2. Conversion routes of lignocellulosic biomass into renewable fuels and chemicals.

3. Electrochemical Hydrogenation (ECH)

Biomass occupies a distinctive role among renewable energy sources, primarily due to its carbon content and capability to convert atmospheric CO₂ into organic matter. This characteristic positions biomass as a promising candidate for sustainable valuable chemicals. Specifically, lignocellulosic biomass, which is derived from plant materials and is inherently renewable, has emerged as a critical focus in the production of bio-jet fuel [28]. Figure 3 provides a schematic representation of the environmentally friendly transformation of biomass-derived compounds, originating from cellulose, hemicellulose, and lignin, into value-added chemicals. This conversion occurs through electrocatalytic hydrogenation within fuel cell systems. In this process, biomass-derived molecules undergo electrochemical hydrogenation in the fuel cell, yielding more valuable and usable chemicals. This approach is consistent with the principles of green chemistry, aiming to diminish dependence on fossil fuels [29–32].

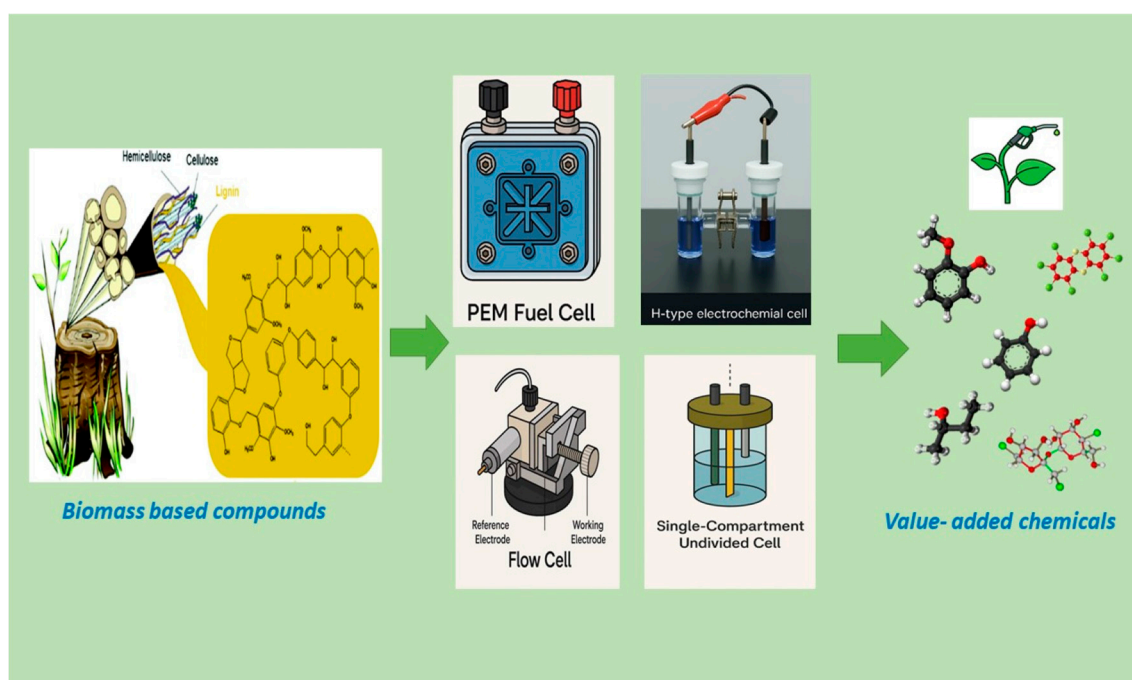


Figure 3. Conversion of biomass-based compounds into high-value chemicals via fuel cell electrocatalysis.

Bio-oils obtained from biomass have properties such as a high oxygen content, acidity, instability, and complex molecular structures that prevent them from being used directly as transportation fuels [33–35]. Researchers are actively working to address these barriers in order to fully harness the potential of biomass through ongoing advancements in technology. Typically, biomass-derived bio-oils are made up of monomeric or dimeric alkoxyphenols, whereas petroleum-based jet fuels are comprised of about 20% n-alkanes, 40% isoalkanes, 20% cyclic alkanes, and 20% aromatic hydrocarbons [36]. Transforming bio-oils into valuable chemicals requires a series of chemical transformations, including deoxygenation, carbon retention, and hydrogenation of unsaturated bonds. Among various methodologies, catalytic hydrodeoxygenation is particularly noteworthy for its ability to produce aromatic hydrocarbons and cycloalkanes with low oxygen content and high stability, thereby improving fuel quality, which is crucial for the promotion of sustainable energy sources. While numerous thermal catalytic hydrogenation techniques have been developed for bio-oil upgrading, they often involve high temperatures and hydrogen pressures. Such extreme conditions can lead to the creation of undesirable byproducts like coke or tar, negatively impacting product quality and system stability [37–39]. Moreover, the catalysts used may become deactivated due to prolonged exposure to elevated temperatures and pressures, thereby reducing their reaction activity and the overall efficiency of bio-oil upgrading. In contrast, electrocatalytic hydrogenation (ECH) offers considerable advantages for modern industrial processes by allowing chemical reactions to take place under milder and more controllable conditions [40–42]. These attributes make ECH a more sustainable and environmentally friendly option. One of the significant benefits of ECH is that it eliminates the need for an external hydrogen supply, which significantly decreases energy consumption and mitigates environmental impacts on both local and global scales. Additionally, the absence of requirements for the storage or transportation of hazardous gases makes ECH a safer, economically viable alternative [43–45].

3.1. The Role of Electrochemical Processes in Sustainable Chemical Conversion and Energy Systems

Electrochemical conversion has a crucial role in achieving sustainable chemical transformations due to its eco-friendly characteristics and significant energy efficiency. By

operating under controlled conditions, these processes enhance selectivity, which allows for the better management of chemical reactions and more efficient system operations. Their ability to reduce fossil fuel consumption makes them integral to strategies aimed at mitigating climate change, particularly through low carbon emissions. The integration of electrochemical technologies into future energy frameworks represents a key advancement for environmental protection [46]. Furthermore, these processes excel in providing high purity and efficiency, often surpassing the capabilities of traditional methods, making them advantageous for industrial and scientific applications. The impact of electrochemical conversion spans various sectors, emphasizing its importance in the development of innovative materials and sustainable energy solutions [47,48].

Figure 4 illustrates the conversion mechanism in an electrochemical cell utilizing a proton-exchange membrane (PEM). In this configuration, electrical energy sourced from renewable resources drives the dissociation of molecular hydrogen (H_2) into protons (H^+), which subsequently interact with organic compounds to produce value-added chemicals. During this process, hydrogen gas is introduced into the cell, where it dissociates into protons as it traverses the PEM. The protons migrate across the membrane while electrons flow through an external circuit. The organic compounds such as phenol or guaiacol undergo electrochemical hydrogenation upon interacting with both protons and electrons, resulting in reduced, more stable, and chemically valuable products. The utilization of renewable energy to power the system further underscores its environmentally friendly nature. As illustrated in Figure 4, the sources of energy and hydrogen, along with the resulting chemical products, are clearly represented.

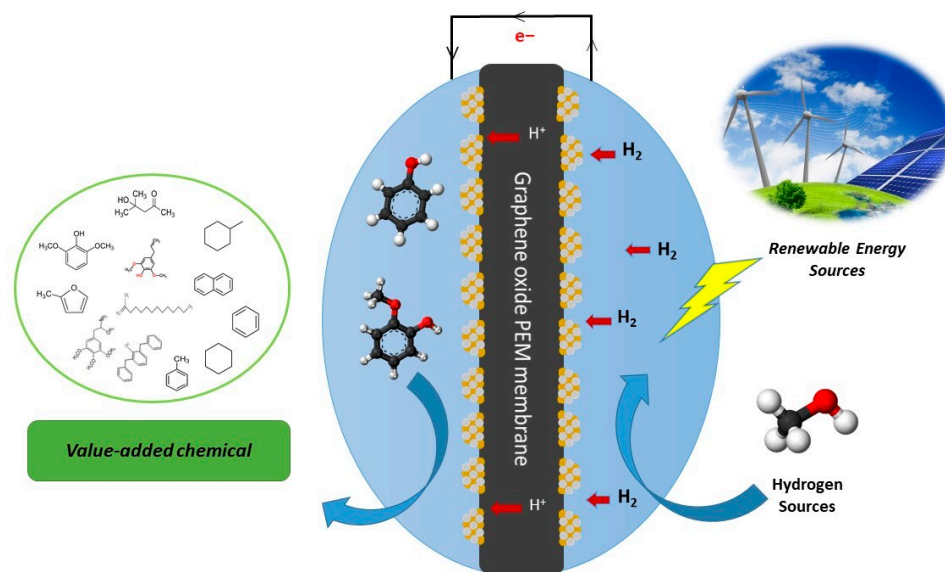


Figure 4. Production of value-added chemicals via electrocatalytic hydrogenation in a PEM cell using renewable energy sources.

3.2. Controlled Selectivity and Low Carbon Emission in Electrochemical Conversion: Advancing Precision and Sustainability in Chemical Processes

The induced selectivity on electrochemical conversion processes is selectively controlled by the material and the applied potential. This unique feature can allow one to preferentially promote desired chemical reactions and at the same time suppress unwanted side production [49,50]. Selectivity in electrochemical systems is due to the tailor-made electrochemical and physical properties of the electrode surface that decelerate transport and reduction or oxidation of one of the two partners in the molecule. By this method, the purity of the products obtained may be significantly increased and the losses by side reac-

tions can be greatly diminished. These enhancements directly serve industrial processes for general improvement and in a cost-effective and in environmentally friendly manner. These developments not only promote the widespread adoption of eco-friendly technologies but also stimulate innovation across emerging industrial sectors. New electrochemical technologies and processes have indeed become strategic in the new industrial approach; this continued study and technological developments will likely result in progressively improved and efficient solutions for this application in the short term [51–53]. These efforts highlight the critical role of electrochemical processes to drive low-carbon industrial efficiency and to advance clean technologies in numerous sectors. Electrochemical process conversion is the key to reaching carbon neutrality, and it is generally recognized to be the ideal way toward an environmentally friendly future energy system [54]. These advanced technologies offer numerous benefits, including the replacement of fossil fuels, a reduction in carbon footprint, and a decreased overall environmental burden [55–57]. In particular, the synergy of electrochemical processes with renewable energies presents a promising opportunity for a substantial reduction in carbon footprint in clean energy generation. As a result, electrochemical technology has become an essential component of pursuing the sustainable future by providing the production of clean and low-carbon emission energy. In addition to their environmental advantages, these techniques are of crucial importance for increasing energy efficiency. Through these two kinds of impacts, electrochemical conversion plays roles in economic and environmental sustainability and helps to lower reliance on fossil fuels.

Recent developments in electrocatalytic hydrogenation (ECH) have brought single-atom catalysts (SACs) to the forefront as a revolutionary category of materials. Their remarkable atom utilization, precisely defined active sites, and impressive electronic adaptability set them apart. According to one study, SACs effectively integrate the principles of homogeneous and heterogeneous catalysis, showcasing enhanced activity and stability in hydrogen-related electrochemical processes, particularly in the hydrogen evolution reaction (HER). The researchers emphasized that SACs achieve complete metal dispersion, optimizing catalytic efficiency, and exhibit robust metal–support interactions that influence charge distribution and the geometric configuration at the active site [58–61].

These catalysts play a vital role in selectively adjusting adsorption energies, such as the hydrogen adsorption free energy (ΔG_{H^*}), which is essential for minimizing unwanted hydrogen evolution reactions (HERs) and promoting electrochemical hydrogenation (ECH) pathways. For example, platinum (Pt) atoms bonded to graphdiyne within a $\text{C}_2\text{-Pt-Cl}_2$ environment demonstrated significantly higher activity compared to other bonding arrangements, achieving an optimal ΔG_{H^*} of 0.092 eV [62]. In another instance, molybdenum (Mo)-based single-atom catalysts (SACs) featuring $\text{W}_1\text{N}_1\text{C}_3$ units recorded a ΔG_{H^*} of 0.033 eV, greatly surpassing other coordination configurations like WN_2C_2 or WN_4 [63]. These findings highlight the direct relationship between the precise coordination of single atoms and enhancements in selectivity and kinetics for catalytic hydrogenation processes.

The study highlights that single-atom catalysts (SACs) utilizing non-precious metals such as cobalt, nickel, iron, and copper are emerging as viable sustainable and economical substitutes for traditional noble-metal-based catalysts [58]. Notably, cobalt single atoms arranged in a $\text{Co}_1\text{P}_1\text{N}_3$ configuration exhibited significantly improved hydrogen evolution reaction (HER) activity when compared to conventional CoN_4 structures [64]. This improvement is attributed to favorable charge distribution and the enhanced stabilization of intermediates. The authors emphasize the importance of appropriate support selection and engineering techniques, which can include heteroatom doping, defect creation, and spatial confinement. These methods are crucial for anchoring single atoms, preventing their agglomeration, and modifying the electronic environment—elements that also

play a significant role in determining electrochemical hydrogen (ECH) selectivity and stability [65–68].

Additionally, SACs can serve as co-catalysts that engage the support structure for activation. For instance, research indicates that Ru atoms integrated within MoS₂ promote the formation of sulfur vacancies and stimulate neighboring S atoms, which results in a reduced water dissociation barrier and ΔG_{H^*} , consequently improving the kinetics of the hydrogen evolution reaction (HER) [69]. Comparable activation effects have been noted in systems such as Pt-doped CoSe₂ and Ni-loaded MoS₂, both exhibiting enhanced charge transfer and favorable hybrid electronic states that facilitate hydrogenation processes [70,71].

Electrochemical conversion technologies are gaining recognition for their potential in promoting sustainable chemical manufacturing, particularly in the area of biomass valorization. A notable method within this field is the direct electrochemical CO₂ reduction (CO₂RR), which effectively transforms CO₂ into C1 products, including carbon monoxide, formic acid, methane, and methanol, under mild conditions [72]. Despite facing obstacles like high overpotential and the competing hydrogen evolution reaction (HER), advancements in catalyst design encompassing single-atom catalysts (SACs), bimetallic nanostructures, and materials derived from metal–organic frameworks (MOFs) have significantly enhanced reaction selectivity and Faradaic efficiency [73]. Additionally, photoelectrochemical (PEC) systems that employ light-activated electrodes have improved CO₂RR performance by facilitating high selectivity through solar-assisted charge transfer. Furthermore, hybrid electro-photo-thermal systems are being developed to address limitations related to mass transfer and reaction kinetics, particularly in multi-electron conversions that yield multi-carbon products. MOFs, characterized by adjustable porosity and active centers, have demonstrated remarkable potential across all three conversion platforms electrochemical, photoelectrochemical, and thermochemical by promoting charge transfer, stabilizing intermediates, and enabling selective adsorption [73]. When powered by renewable electricity, these technologies present a promising low-carbon approach to generating value-added chemicals and fuels, thereby supporting global climate and energy objectives.

The results demonstrate that single-atom catalysts (SACs), especially those utilizing non-precious metals, are significantly important for contemporary electrochemical hydrogenation (ECH) systems. Incorporating this information into the introduction and discussion parts of the manuscript will improve its scientific relevance and ensure it is consistent with current frameworks in catalyst development.

3.3. Electrochemical Conversion in PEM Fuel Cells: Distinct Physical and Chemical Advantages

Electrochemical conversion processes in fuel cells utilizing proton-exchange membranes (PEMs) demonstrate distinct physical and chemical characteristics compared to conventional methods. Physically, these operations typically occur under lower temperature and pressure conditions, leading to reduced energy consumption and enhanced overall process efficiency. Chemically, reactions in PEM systems involve proton transport through the membrane, occurring near the electrode surface, which optimizes reaction selectivity [74,75]. By carefully adjusting electrode materials and applied electrical potentials, specific hydrogen and oxygen products can be selectively generated while minimizing undesired byproducts. Additionally, electrochemical processes in PEM fuel cells generally result in lower carbon emissions, contributing to environmental sustainability [76].

The electrochemical reaction mechanisms are influenced by electrode potential as well as the intrinsic properties of the electrode surface, particularly the effectiveness of the catalyst. The catalyst significantly impacts the adsorption of electroactive species,

electron transfer rates, and overall reaction efficiency, all of which depend on the pH of the surrounding environment [76]. The chemical nature of the reactants and products further shapes the reaction pathway, necessitating careful consideration of catalyst structure and composition during electrode and electrolyte optimization. These considerations are vital for ensuring safety and maximizing system efficiency. Ultimately, factors such as electrode potential, catalyst properties, and material optimization are crucial for the performance of PEM fuel cells [77–80]. Electrochemical hydrogenation involves saturating unsaturated organic compounds using electrode-bound hydrogen. This method operates under a much lower temperature and pressure than conventional hydrogenation, offering advantages such as reduced energy consumption and lower carbon emissions [81].

The hydrogenation mechanisms include complex sequences of proton and electron transfer steps, with electrode material and applied potential being critical for selectivity and kinetics. Furthermore, electrocatalyst surface characteristics and proton mobility are key to mechanism effectiveness. Ongoing research aims to enhance efficiency and environmental friendliness in line with green chemistry principles [82,83]. Electrode and membrane materials are vital in electrochemical conversion processes. Electrode materials influence reaction kinetics and efficiency, while membranes regulate ion transfer and affect overall system performance. Advanced materials like graphene-based membranes, PEMs, and ion-exchange membranes are recognized for their high conductivity, selectivity, and stability, which enhance application efficiency [84–86]. Metal–organic frameworks (MOFs) offer additional benefits in gas storage and separation due to their porous structures, making the careful selection of these materials crucial for optimal energy efficiency in electrochemical systems [86–88]. Graphene-based membranes, widely used in electrochemical applications, possess a high surface area, mechanical strength, and excellent conductivity. Their controlled layer arrangement provides high ionic conductivity and enhanced selectivity, proving effective in critical tasks like gas separation and energy storage [89]. Furthermore, these membranes control the passage of potentially contaminating species during electrochemical reactions, contributing to improved process efficiency. Their ability to enhance energy efficiency underscores their potential in environmentally friendly technologies that promote public health and sustainability.

3.4. Proton-Exchange Membranes (PEMs) and Their Functional Role in Electrochemical Devices

Proton-exchange membranes (PEMs) are specialized polymeric materials meticulously engineered to facilitate the transport of protons. These membranes are crucial in various electrochemical applications, especially in fuel cells. PEMs possess high proton conductivity, enabling efficient proton transport while preventing the unintended mixing of fuel and oxidant. This dual capability significantly enhances energy conversion efficiency and contributes to overall energy conservation [90]. Constructed from unique polymers that incorporate sulfonic acid groups, PEMs exhibit high proton conductivity while maintaining low electronic conductivity, thereby reducing energy losses. Additionally, their exceptional resistance to fluctuations in temperature and humidity underpins their capacity for long-term, reliable performance. As such, PEMs are invaluable in the advancement of alternative energy sources that aim to mitigate reliance on fossil fuels [91,92]. Figure 2 illustrates the structure of a PEM fuel cell that integrates a graphene oxide (GO)-based proton-conducting membrane.

This multilayered design enhances electrochemical reaction efficiency by integrating mechanical support, gas diffusion, and proton conduction functionalities. The Teflon plate provides mechanical support and ensures environmental isolation, thanks to its chemical inertness, which prevents undesirable reactions with active chemical species. The carbon paper layer, positioned between the electrode and the membrane, acts as a

gas diffusion layer, promoting the homogeneous distribution of gases, namely hydrogen and oxygen, across the active surface while supporting electrical conductivity. The carbon electrode layer is the site of electrochemical reactions, wherein electrocatalyst-supported electrodes facilitate effective electron transfer and enhance interactions between reactive gases (H_2 and O_2) and the membrane. Central to this system is the graphene oxide membrane, which functions as the primary proton-conductive element. Its superior proton conductivity allows for the efficient transfer of H^+ ions from the anode to the cathode. In contrast to traditional Nafion membranes, GO membranes present a more sustainable and economically viable alternative from an environmental standpoint. Each layer in this assembly critically impacts both the operational efficiency and long-term stability of the fuel cell. Figure 5 portrays a well-structured and functional PEM cell, encompassing these essential components.

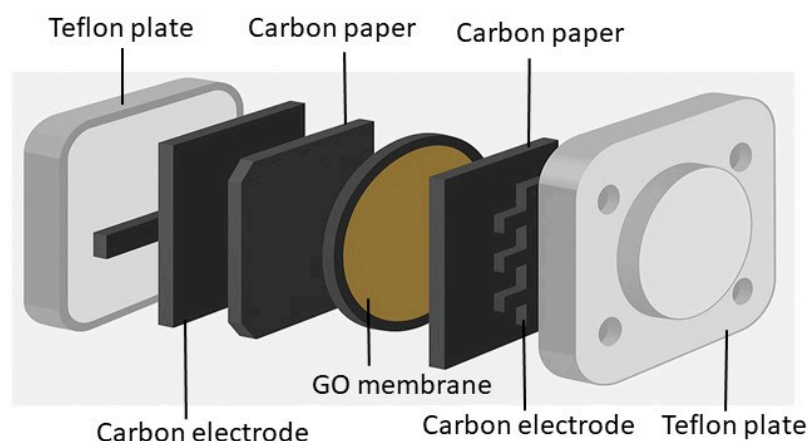


Figure 5. Schematic diagram of a GO membrane-based electrochemical cell.

Graphene oxide (GO)-derived membranes have stimulated intense interest in the realm of electrochemical applications due to their outstanding ionic conductive nature, mechanical stability, and tunable barriers. Compared to conventional membranes, such as the pristine Nafion[®] (Chemours Company, Wilmington, NC, USA), polyvinyl alcohol (PVA), or that containing Vulcan XC-72, GO membranes often have superior selectivity, favorable crossover behavior, and comparable conductivity. Small organic molecules' and gases' crossover can also be effectively suppressed in GO membranes. For example, treating Nafion membranes with GO/DGO interlayers showed a remarkable decrease of 55% in organic crossover and initial hydrogen permeability decrease, followed by a 12% increase in open circuit voltage (OCV) in one of the studies [93,94]. These findings were accompanied by a temporary increase in HFR, the magnitude of which later settled down with the restacking of GO layers, providing long-term reproducibility. Pt nanoparticles supported on reduced graphene oxide (Pt/rGO) were found to be superior to the Pt/C catalyst in the electrochemical hydrogenation of soybean oil as compared to Vulcan XC-72-based carbon electrodes. With the Pt/rGO system, the efficiency and distribution of the catalyst are improved, and the content of TFAs was also remarkably decreased to 1.53% (9.8% for traditional processes: [95]). This improvement is due to the large S/B ratio and high electrical conductivity of the GO matrix that promotes efficient electron transfer and stability. Another study was conducted and compared GO-doped chitosan/PVA membranes with commercial anion-exchange FAA and Neosepta. The observed higher hydroxide ion conductivity ($0.379\text{ mS}\cdot\text{cm}^{-1}$ than $0.253\text{ mS}\cdot\text{cm}^{-1}$) (GO-modified membrane) and much lower permeability of the alcohol qualifies them for applications in alkaline electrochemical reactors [96–98]. Furthermore, the GO sheet increased the mechanical and thermal stability of the polymer matrix by means of hydrogen bonding, ensuring long-term

durability. In addition, the water dispersibility of GO-based membranes allows them to be used in eco-friendly polymer systems (e.g., chitosan, PVA), offering sustainable membrane preparation procedures [99].

Research reveals that the GO-based membrane reactor can provide effective proton transfer, which is able to realize 95% deuterium incorporation and a 90% yield of ethynylbenzene at room temperature and atmospheric pressure. These values are comparable to or even greater than those obtained with conventional Nafion membranes. The enhanced electrocatalytic activity of GO membranes is attributed mainly to the oxygen-enriched functional groups on the GO surface which favor the dispersion of the catalyst and improve ion transport. By comparison, control experiments not using GO membranes or a catalyst resulted in 5% deuteration, highlighting the necessary role of GO in the overall efficiency of the system [100]. Additionally, the GO membrane has showed great mechanical and operation stability for long-term use and has been implemented on various substrates including alkynes, aldehydes, arenes, and pharmaceuticals with high yields. These results suggest that GO membranes not only exhibit comparable conductivity with commercial materials such as Nafion but also offer enhanced micromolecular selectivity and operational flexibility for electrocatalytic application.

Graphene oxide-based membranes have been widely recognized as a promising material in electrochemical devices for the enhancement of multiple functions. The oxygen atoms present in the rich functional groups such as carboxyl, epoxide, and hydroxyl groups provide an easy path for the ions to transport effectively and this consequently results in an increased conductivity for protons or hydroxide ions. Furthermore, the GO inside the membrane structure forms a selective physical barrier that greatly decreases the unnecessary permeation of hydrogen and small organic molecules, which is particularly important in improving the reactor's performance. In addition, the GO nanosheets form good interactions with the polymer backbone by hydrogen bonding which improves their thermal stability and mechanical properties. On the microstructure, the sp^2 -hybridized wired network of GO is characterized by a large specific surface area and excellent electrical conductivity, which can induce the uniform distribution and activation of the supported catalysts. This enhanced dispersion is crucial for improving EC activity and is also significantly useful in suppressing the undesired trans fatty acids generation during hydrogenation reaction, as the matrix of GO can regulate the reaction pathway and selectivity actively.

3.5. Ion-Exchange Membranes in Electrochemical Systems: Functionality, Types, and Applications

Ion-exchange membranes are a kind of special membrane material which can realize selective ion transfer between solutions in an electrochemical system. Functioning based on the ability to facilitate cation or anion exchange, these membranes are widely used in various applications such as water purification, energy storage, and electrodialysis. The ion selectivity and conductivity of ion-exchange membranes are abstracted as a few key parameters in order to improve several electrochemical processes' efficiency and effectiveness [101]. Moreover, the chemical and thermal stability inherent in these membranes underpins their long-term performance, which is integral to the sustainability of operational processes. The ion-exchange capacity of ion-exchange membranes is influenced by the specific type and structure of the polymer matrix from which they are fabricated. This diversity allows for the development of application-specific solutions that are optimized to meet varying technological requirements [102,103]. Figure 3 provides a schematic representation of the three primary types of ion-exchange membranes: anion-exchange membranes, cation-exchange membranes, and proton-exchange membranes (PEMs). Each type operates based on the selective permeability of electrically charged ions and plays a crucial role in various electrochemical applications. Anion-exchange membranes are equipped with

fixed positively charged groups that permit the passage of negatively charged ions (anions), while simultaneously repelling positively charged ions (cations). For example, these membranes are employed in electrochemical systems to facilitate the transport of hydroxide ions (OH^-). In the accompanying diagram, the flow of anions through the membrane is illustrated with a blue arrow, while the cations are depicted as blocked.

In contrast, cation-exchange membranes comprise fixed negatively charged groups that permit the passage of positively charged ions (cations) and obstruct the transport of anions. As the illustration suggests, cations penetrate the membrane while anions do not. These membranes are frequently used in desalination and ion-separation treatments. The so-called proton-exchange membranes (PEMs), a sub-group of cation-exchange membranes, are mostly used in fuel cells. The side chains have fixed negative charges that can conduct only protons (H^+) and repel other ions. This preferential ionic conduction is needed in PEM fuel cells to control the movement of H^+ from the anode to the cathode to avoid losing efficiency. Figure 6 shows the charge features and ion selectivity for each membrane, explaining clearly the basic mechanisms on the ion-exchange processes and their operational mode.

Types of Ion Exchange Membranes

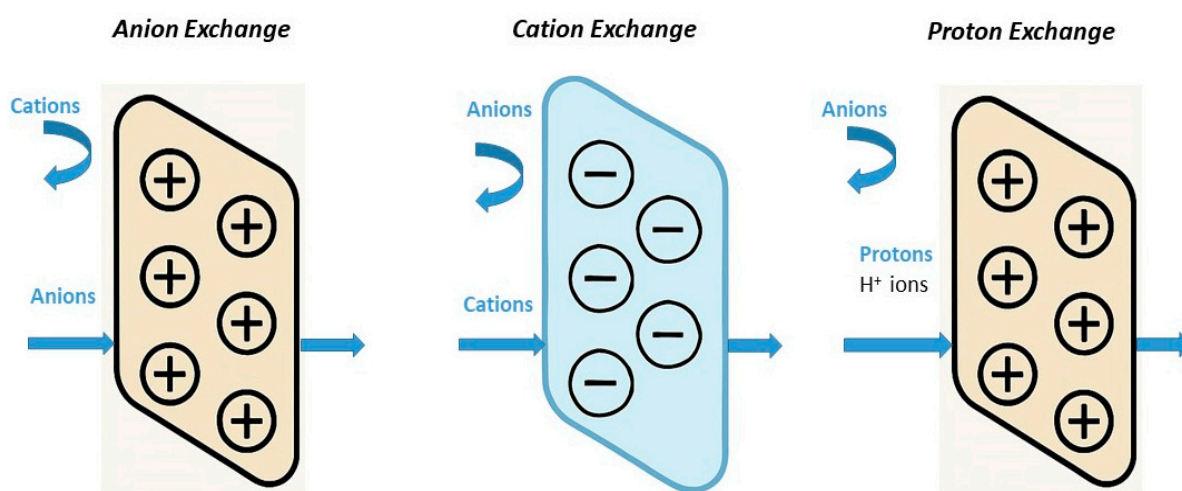


Figure 6. Types and transport mechanisms of ion-exchange membranes.

3.6. Metal–Organic Frameworks (MOFs): Structural Advantages and Applications in Electrochemical Systems

Metal–organic frameworks (MOFs) represent an intriguing category of crystalline materials characterized by the coordination of metal ions with organic ligands, which results in distinctive structural attributes. Their remarkable porosity and a wide range of structural configurations render them particularly advantageous for various electrochemical applications across multiple domains. With significant surface areas and highly adaptable pore sizes, MOFs demonstrate substantial potential in key processes such as gas storage, separation, and catalysis. The high functional efficiency and customizable features of MOFs have led to increasing scholarly interest due to their contributions to technologies focused on energy storage, environmental remediation, and chemical transformations, underscoring their importance in contemporary scientific inquiry and industrial use [104,105].

One of the most notable characteristics of MOFs is their exceptional capacity for modification tailored to specific application requirements. This adaptability is especially beneficial for processes that necessitate both low energy consumption and high opera-

tional selectivity. By utilizing innovative combinations of various metals and organic ligands, researchers can effectively refine MOFs for enhanced capabilities across diverse electrochemical systems. This optimization is pivotal in fostering advancements in sustainable energy initiatives, positioning MOFs as critical components in the development of cutting-edge energy technologies.

As illustrated in Figure 7, MOFs have been embedded into electrochemical hydrogenation (ECH) systems, serving as active materials at the cathode. Their high surface area and adjustable pore sizes promote the selective reduction of target compounds on the electrode's surface. Additionally, the integration of MOF-derived structures and MOF-based composites markedly improves both conductivity and structural integrity. These enhancements lead to a significant increase in the overall efficiency of the ECH process. Thus, MOF-based materials represent a promising pathway for sustainable fuel generation, accentuating their vital role in the development of energy solutions that prioritize efficiency and environmental sustainability.

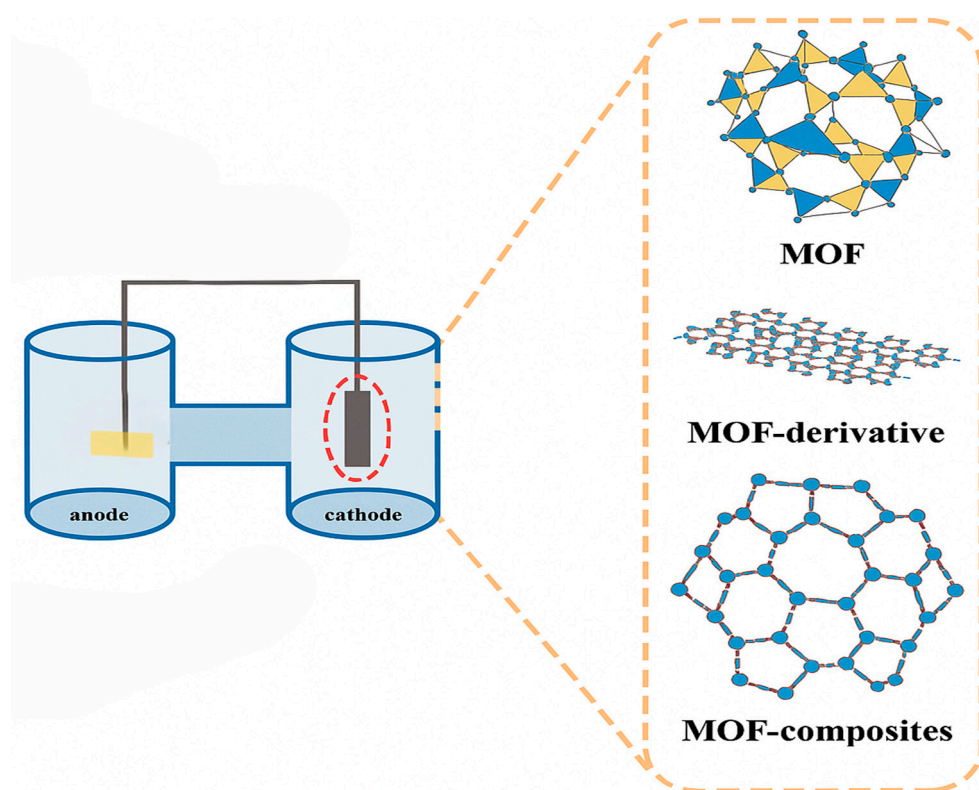


Figure 7. Schematic representation of MOF-based materials in electrochemical hydrogenation systems.

4. Electrochemical Hydrogenation of Biomass-Derived Compounds Using Various Electrocatalysts, Membranes, and Electrode Systems

Consequently, the capacity for researchers and engineers to efficiently investigate the potential of metal–organic frameworks (MOFs) has significantly increased, allowing for an expanded range of applications in electrochemical and energy-related domains [106]. In this context, Table 1 summarizes various studies on electrocatalytic hydrogenation that were carried out using different membranes and catalysts. This table emphasizes the pivotal role that MOF-based systems play in these electrocatalytic processes.

Table 1. Electrochemical hydrogenation of biomass-derived compounds.

Substrate	Catalyst/Electrocatalyst/Electrode	Membrane	Products	Ref.
Phenol	Pt/Pt, Pt/C, Pd/C, Ru/C, Rh/C Rh/Ni, Ru/Ni, RaNi, Pt alloys: Cr, V, Co, Ir	Nafion-324	Cyclohexanol, Cyclohexanone	[107–111]
	Pt/G, Rh/G, Pd/G, Rh/C, Ru/TiO ₂ , Pt/SSB, RVC-Pt mesh, RVC-graphite, Pt ₃ RuSn/CC, Pt ₁ Rh ₁ /MCN	Nafion-117	Cyclohexanol, Cyclohexanone, Cyclohexane	[112–119]
Glucose	Pb(Hg), Zn,	Nafion, type XR 475	Sorbitol	[120,121]
	RaNi,	Nafion-324	Sorbitol	[122]
5-Hydroxymethyl Furfural (5-HMF)	Cu	Nafion-117	2,5-dimethylfuran (DMF), 2,5-di (hy- droxymethyl)furan, 5-methylfurfuryl alcohol, and 5-methylfuran-2- carbaldehyde.	[123]
Guaiacol	Ru/ACC, Pt gauze, Ti gauze, Ni gauze, PtRhAu	Nafion-117	Phenol, 2- Methoxycyclohexanol, 2- Methoxycyclohexanone	[124–127]
	Ra-Ni	Nafion-117	Phenol, 3- Methoxycyclohexanol, 4- Methoxycyclohexanol	[128]
	Rh/ACC	Nafion	Cyclohexanol	[129]
	RhPtRu	Bipolar membrane	Methoxy- cyclohexanes	[130]
Levulinic acid (LVA)	Pb, Cu, Fe, Ni, C	Nafion-117 Fumasep [®] FKE (FuMA-Tech GmbH, Ludwigsburg, Germany)	Valeric acid, γ -valerolactone	[131–133]
Furfural	Pt, Pb, Ni, Cu, Pt/ACF, C, Fe, Ti/La nano-TiO ₂ , Ti/nano-TiO ₂ , Rh/C, Pt/C, Pd/C,	Nafion Nafion-117 Nafion-115 Fumasep [®] FKE	Furfuryl alcohol, methyl furan, 2-methylfuran, Pinacol, furoic acid, tetrahydrofurfuryl alcohol, 2- methyltetrahydrofuran	[42,134–137]
Phenolic compounds	PtNiB/CMK-3, Graphite rod	Nafion-117	Cyclohexanol, cyclohexanone, Cyclohexane	[138,139]

A systematic investigation into the influence of the electrochemically active surface area (ECSA) on the electrocatalytic hydrogenation of phenol found that current efficiency remained stable for Pt loadings of 2% to 30%, but activity decreased at 60% Pt content, indicating the reaction is structure-sensitive. The study evaluated various electrode materials and alloy compositions regarding both current efficiency and product selectivity. Among the tested electrodes, Pt/C had the highest electrocatalytic activity, while Ru/C

showed the highest selectivity for cyclohexanol. In contrast, Pd/C had the lowest efficiency and selectivity. The Pt–Co/C alloy matched the activity of Pt/C with improved cyclohexanol selectivity, while Pt–Ir/C and Pt–Rh/C displayed similar metrics. These findings highlight the importance of the electrode material and alloy composition in optimizing electrocatalytic performance [109].

A study examined the electrocatalytic hydrogenation of phenol with carbon-supported highly dispersed Pt (Pt/C) electrodes. Comparing Pt/C and Pt/Pt electrodes with equal Pt amounts or similar electrochemical surface areas showed that Pt/C electrodes had a higher catalytic activity. This was due to a larger surface area and beneficial metal–support interactions. Mechanistic analysis revealed the rate-determining step involved surface reactions between adsorbed species, not the adsorption of phenol, applicable to both electrode types. These findings highlight the importance of support materials and interactions in electrocatalytic activity, establishing Pt/C as an effective electrocatalyst for phenol hydrogenation and related reactions [110].

The process of thermal catalytic hydrogenation (TCH) and electrocatalytic hydrogenation (ECH) of phenol and benzaldehyde using platinum group metals such as platinum (Pt), palladium (Pd), and rhodium (Rh) has been studied. Because reactions in the aqueous phase are not well understood, the study involved extensive kinetic modeling in addition to experimental work. A Langmuir–Hinshelwood-type model for aqueous systems was used and determined the most rate-determining process to be the reaction of surface-adsorbed hydrogen (H) with organics. The hydrogenation process is characterized by the addition of H* to the adsorbed species of phenol or benzaldehyde, with competition for surface sites significantly affecting reaction rates. The adsorption equilibrium constants determined for Pt/C electrodes corresponded to those observed for Pt(111)-like surfaces, suggesting a high reactivity level. Palladium exhibited distinctive behavior during the hydrogenation of benzaldehyde; strong adsorption on Pd led to reduced hydrogen coverage, which in turn improved the Faradaic efficiency. Conversely, in the hydrogenation of phenol, Pd transitioned to a β -PdH_x phase, which impeded the reaction kinetics. Under ECH conditions, an increase in cathodic potential was found to elevate hydrogen density, effectively reducing activation energy and elucidating the kinetic advantages associated with ECH. Palladium demonstrated notable selectivity for the hydrogenation of benzaldehyde, whereas the Faradaic efficiencies of platinum and rhodium were lower, attributed to variations in surface coverage and reactivity. This analysis provided valuable insights into the mechanisms and pathways of ECH and TCH, underscoring the critical rate-limiting steps and surface characteristics essential for the enhancement of electrode materials in the selective hydrogenation of phenol and benzaldehyde [111].

A recent study introduced a method for the electrocatalytic hydrogenation (ECH) of phenol using Pd/ γ -Al₂O₃ catalysts with 10% Pd. It evaluated how solvent composition and electrolyte characteristics affect ECH efficiency. Catalyst powders were mixed in aqueous solutions with methanol and aliphatic carboxylic acids (acetic, propionic, and butyric) and passed through an RVC cathode. The results showed that the ECH efficiency decreased with higher methanol concentrations but improved with longer carbon chains of aliphatic acids. A correlation was found between phenol adsorption on the Pd–alumina surface and ECH efficiency via dynamic adsorption isotherms. The study demonstrated in situ functionalized materials for ECH, showing that aliphatic acids attach to the alumina surface via carboxylate groups, affecting the catalyst's properties. DRIFT spectroscopy and temperature-programmed mass spectrometry confirmed the stable adsorption of aliphatic acids below 200 °C. This methodology enables the regioselective hydrogenation of phenol and more intricate organic compounds, enhancing efficiency and selectivity in ECH through the in situ surface modification of Pd–alumina catalysts [112].

A recent investigation studied the electrocatalytic hydrogenation (ECH) of phenol into cyclohexane and cyclohexanol using an H-type electrochemical cell with a platinum plate anode. The cathode choice was vital for ECH efficiency, with a 1.5% Pt-supported graphite electrode performing best. At 20 °C, the cyclohexane yield was 44.1%, rising to 63.7% at 60 °C but falling to 50.3% at 80 °C, suggesting moderate temperatures optimize efficiency. A 0.2 mol/L HClO₄ solution proved the most effective electrolyte. Increasing the current from 10 to 30 mA boosted yield, but levels over 70 mA decreased it due to side reactions, including hydrogen gas production. The optimal phenol concentration for peak yield was 50 mmol/L. This study indicates Pt/graphite electrodes are promising for the low-temperature hydrogenation of phenols, potentially aiding in energy-efficient biofuel production. Controlled ECH under refined conditions may enhance sustainable fuel synthesis and further research on other phenolic substances [113].

A recent study compared electrocatalytic hydrogenation (ECH) and mild thermal hydrogenation (TH) of phenol using metal-loaded Pt/C, Rh/C, and Pd/C catalysts in electrochemical cells. It evaluated factors like electrolyte type, pH, current density, temperature, and catalyst concentration on ECH performance. Rh/C had the highest turnover frequency (TOF), with Pt/C also active, while Pd/C showed lower conversion and selectivity. The ECH mechanism, identified as Langmuir–Hinshelwood-type, involved adsorbed H⁺ ions and phenol, with H• radicals hydrogenating phenol, unlike TH, which required dissociative H₂ adsorption. The activation energy was 30 kJ/mol for Pt/C and 23 kJ/mol for Rh/C in ECH. ECH's performance was less affected by hydrogen gas formation, indicating independent operation from TH. Both methods followed the phenol–cyclohexanone–cyclohexanol sequence without C–O bond cleavage or dehydration, emphasizing environmental factors acting on adsorption and proton reduction. The study suggests that the low-temperature ECH of phenolic compounds can match the activation energies of thermal methods but shows different kinetics, highlighting Rh/C and Pt/C as effective electrocatalysts for sustainable chemicals and the necessity of controlling hydrogen gas to enhance ECH selectivity and efficiency [114].

In an investigation, the electrocatalytic hydrogenation (ECH) and thermal catalytic hydrogenation (TCH) of phenol were studied using Pt and Rh catalysts, focusing on temperature effects. Both rates increased with temperature up to 60 °C but decreased afterward due to surface poisoning from dehydrogenated phenol derivatives. Higher temperatures and lower H₂ pressures worsened poisoning, leading to significant deactivation. In both methods, a higher potential in ECH or H₂ pressure in TCH improved surface hydrogen coverage (H*), thus increasing reaction rates. Sufficient H* allowed the removal of surface poisons, resulting in a 130-fold TCH rate increase at 80 °C under high pressure. The rate-determining step involved adsorbed hydrogen (H*) reacting with phenol, with an activation energy of ~33–34 kJ/mol. The ECH rates were lower than TCH, due to less H* coverage and higher poisoning susceptibility. At higher conversions, cyclohexanone competed for adsorption, decreasing the rate. The study proposed a microkinetic model for both mechanisms to reduce surface poisoning, highlighting TCH's superior performance under high pressure and suggesting improvements for ECH catalysts to enhance rehydrogenation, vital for low-temperature hydrogenation of phenolic compounds [115].

This study introduces a new method for the electrocatalytic hydrogenation (ECH) of bio-oil compounds by developing effective electrocatalysts and identifying active sites. Shrimp shell biochar (SSB) was used for the first time as a support for Pt electrocatalysts. The Pt/SSB catalyst achieved 100% conversion of phenol, with 98% selectivity for cyclohexanone and cyclohexanol in 5 h. It also demonstrated stability, retaining 94.3% conversion over three cycles without a notable loss in selectivity. The hydrogenation begins with the reaction of C=C bonds in phenol with hydrogen species, leading to cyclohexanone,

then cyclohexanol, with some phenol directly forming cyclohexanol through simultaneous hydrogenation. Analyses revealed that Pt, Pt–Nx, and C=O sites are vital for phenol's hydrogenation, with Pt–Nx sites mainly contributing to selectivity for cyclohexanol. This research highlights the potential of nitrogen-rich biochar-supported Pt catalysts for upgrading biomass-derived bio-oils, with SSB serving as a low-cost support material. Identifying active sites aids in designing advanced ECH catalysts for converting bio-oils into valuable fuels and chemicals [116].

A highly efficient PtRh bimetallic catalyst was developed for the electrocatalytic hydrogenation of phenol under acidic conditions, aiming to optimize adsorbed hydrogen (H_{ad}) usage and enhance reaction selectivity against hydrogen evolution reaction (HER). PtRh bimetallic nanoparticles were uniformly dispersed on mesoporous carbon nanospheres (MCNs), improving effective hydrogen usage in the H–UPD region (potential > 0 V vs. RHE). The strong interaction between Pt and Rh promoted phenol adsorption, while Rh's electron modification reduced H_{ad} interaction, increasing the Faradaic efficiency (FE) by favoring hydrogenation over HER. DFT calculations outlined two pathways for electrocatalytic hydrogenation leading to cyclohexanone and cyclohexanol via $*C_6H_{10}OH$. The PtRh alloy exhibited optimal adsorption energy for these intermediates, achieving a Faradaic efficiency of 86% with high intrinsic activity at -0.02 V (vs. RHE). The equilibrium between hydrogen and phenol surface coverage was crucial for ECH efficiency, with alloy engineering managing this balance effectively. This study validates bimetallic alloy design as a means to enhance ECH performance, suppressing HER to improve energy efficiency for biofuel and chemical production, paving the way for integrated electrochemical systems for sustainable applications [117].

A recent study improved the design of multi-metallic electrocatalysts for hydrogenating lignin-derived phenolic compounds. Mono-, bi-, and tri-metallic nanoparticles of Pt, Ru, and Sn were supported on carbon cloth to convert phenol into cyclohexanol. The Pt_3RuSn/CC trimetallic electrode displayed a remarkable performance, achieving 91.5% phenol conversion and 96.8% selectivity for cyclohexanol, with low cyclohexanone production. DFT calculations indicated that the synergy of Pt and Ru enhanced the activation of phenolic rings, while Sn contributed additional adsorption sites strongly interacting with carbonyl oxygen, facilitating the conversion of cyclohexanone to cyclohexanol. This trimetallic system outperformed monometallic catalysts like Pt or Ru. Performance was ranked: $Pt_3Ru/CC > Pt_3RuSn/CC > Pt/CC > Pt_3Sn/CC > Ru/CC > Ru_3Sn/CC$. Activation energies were recorded at 20.37 ± 2.04 kJ/mol for Pt/CC and 12.06 ± 2.15 kJ/mol for Pt_3Ru/CC , showcasing Ru's kinetic advantage. Linear sweep voltammetry (LSV) demonstrated the correlation between peak shifts and catalytic activity after phenol addition, indicating that electrochemical measurements can predict catalytic performance. This research provides insights into developing synergistic, multi-metallic electrocatalysts for selective biomass-derived compound hydrogenation, highlighting Sn's role in creating active sites and enhancing understanding of active site functions through DFT [118].

This study presents a novel electrocatalytic hydrogenation (ECH) method for treating phenol-laden wastewater sustainably while recovering valuable products. ECH enhances transformation speed and selectivity compared to traditional electrooxidation (EO), offering scientific and economic benefits. The three-dimensional Ru/TiO₂ electrode demonstrated outstanding ECH performance due to its large active surface area, enabling rapid conversion of phenol to cyclohexanol with a pseudo-first-order rate constant of 0.135 min^{-1} for 1 mM phenol, which is 34-times more efficient than EO. The system remained effective across a pH range of 3–11 and stable at high phenol concentrations (1000 mg/L). Functional groups like chlorine or methyl had a minimal impact on efficiency. This method ensures pollutant removal while recovering high-value chemicals without additional chemicals, pro-

ducing valuable, low-toxicity cyclohexanol. It aligns with zero-carbon emission principles and promotes low-carbon objectives. ECH provides a sustainable solution for eco-friendly conversion of phenolic wastewater, yielding clean water and valuable chemicals with significant potential for advancing wastewater treatment and resource recovery [119].

A recent study examined paired electrochemical synthesis of organic chemicals, revealing up to 50% energy savings compared to conventional methods. In a single cell, glucose was oxidized to gluconic acid at the anode and reduced to sorbitol at the cathode: Anode: Glucose $\rightarrow$ Gluconic Acid (oxidation); Cathode: Glucose $\rightarrow$ Sorbitol (reduction). This concurrent process in one electrolytic cell enhances energy efficiency with dual product output. The setup utilized a zinc cathode, graphite anode, glucose concentration of $0.8 \text{ mol}\cdot\text{dm}^{-3}$, NaBr electrolyte at $0.8 \text{ mol}\cdot\text{dm}^{-3}$, pH of 7, and flow rate of $0.81 \text{ L}\cdot\text{min}^{-1}$. Product yields were 26% for sorbitol and 68% for gluconic acid, with overall yields of 34% and 47% (at 0.4 F/mol). The theoretical yield for each reaction is 50%. Key inefficiencies stemmed from the hydrogen evolution reaction and re-reduction of intermediates. This work highlighted that bidirectional electrochemical conversion conserves energy and allows the concurrent synthesis of valuable products. Sorbitol is a sweetener, while gluconic acid serves as a food additive and metal chelator, aligning with sustainable chemistry goals by promoting energy recovery and multiproduct synthesis [121].

A recent study explored the electrocatalytic hydrogenation (ECH) of guaiacol, presenting a green method to transform lignin-derived compounds into valuable, low-toxicity, water-soluble products like cyclohexanol. The results indicated that reaction conditions and catalyst structure greatly impact product selectivity and Faradaic efficiency. Reactions were conducted in a membrane-separated, stirred-slurry electrochemical reactor (SSER) with a 5 wt% Pt/C catalyst at 1 atm pressure and temperatures between 25 and 60 °C, utilizing Nafion 117 membranes. Several catholyte–anolyte combinations, including H_2SO_4 , NaCl, and NaOH, were assessed. The best efficiency emerged from acid–acid and neutral–acid combinations; notably, a NaCl (0.2 M) and H_2SO_4 (0.2 M) mixture achieved 70% Faradaic efficiency and favorable selectivity for cyclohexanol after 4 h at 50 °C and $109 \text{ mA}/\text{cm}^2$ current density. This was attributed to efficient proton transport across the membrane. A basic medium (NaOH) caused guaiacol deprotonation, decreasing catalytic activity. The process involved guaiacol hydrogenation via adsorbed hydrogen radicals (Hads). Under moderate conditions, cyclohexanol and 2-methoxycyclohexanol were the main products, while cyclohexanone was favored at higher temperatures and current densities due to shifts in Hads coverage. Product selectivity depended on proton concentration, temperature, and cathode potential. The SSER system benefits included effective catalyst suspension, enhanced mass and heat transfer, high current densities (up to $255 \text{ mA}/\text{cm}^2$), low Pt loading efficiency (guaiacol/Pt ratio of 315), and excellent catalyst reusability. However, challenges such as catalyst wear, electrical contact issues, and recovery arose. This study underscored the significance of optimizing conditions, proton transport, and surface chemistry for efficient and selective guaiacol ECH, supporting the sustainable conversion of lignin and showcasing SSER systems' industrial potential [125].

This study explores converting biomass-derived phenolic compounds, like phenol and guaiacol, into valuable chemicals via electrocatalytic hydrogenation/hydrogenolysis (ECH). It assesses various carbon-supported metal catalysts (Pt/C, Ru/C, Pd/C; 5% metal loading) and the influence of different electrolytes (acid–acid and neutral–acid) on reaction selectivity, activity, and Faradaic efficiency. Reactions took place in a membrane-separated stirred-slurry electrochemical reactor (SSER) using a Nafion 117 membrane. At a fixed surface current density ($-109 \text{ mA}\cdot\text{cm}^{-2}$) and 50 °C, Pt/C exhibited the highest activity for the ECH of phenol and guaiacol. Ru/C was notably more active in neutral–acid electrolytes, particularly at higher pH (9–11), suggesting that hydroxide ions enhance Ru's

reactivity. Guaiacol conversion mainly produced 2-methoxycyclohexanol through aromatic ring saturation, while Pt/C in NaCl–H₂SO₄ led to cyclohexanol and cyclohexanone via demethoxylation. This underscores the importance of metal surface–electrolyte interactions on conversion rates and selectivity. Factors like electrode potential, pH, and hydrogen radical surface coverage affect the ECH mechanism, achieving high Faradaic efficiencies. The SSER system's efficient mass/heat transfer makes it suitable for industrial applications. This research outlines catalyst selection and electrolyte design, highlighting the activity retention of Pt/C catalysts post-filtration, emphasizing their practical significance for electrochemical reactor design [126].

This study presents a novel method to produce methoxylated cycloalkane derivatives from lignin-based monomers. Previous efforts to selectively hydrogenate lignin-derived methoxylated compounds faced challenges due to unwanted reductions of the methoxy (–OCH₃) group. The new PtRhAu electrocatalysts allow for the selective conversion of lignin monomers such as guaiacol into valuable products like 2-methoxycyclohexanol (2MC), demonstrating high selectivity. The alloy enhances the surface electronic structure of Pt, improving guaiacol adsorption and hydrogenation while preventing demethoxylation. X-ray absorption spectroscopy (XAS) and in situ Raman studies indicate that Rh and Au contribute to electron donation to Pt, optimizing the surface energy. The catalyst achieved recorded 2MC production with a Faradaic efficiency of 58% and a partial current density of 116 mA·cm^{−2}, surpassing previous benchmarks. Density functional theory analyses and electroanalytical data suggest improvements in reaction energetics, increasing guaiacol coverage and avoiding –OCH₃ cleavage, essential for selectively producing methoxylated products. The study also emphasizes an integrated lignin biorefinery system that converts wood-derived lignin monomers into pharmaceuticals, demonstrating effectiveness with syringyl-rich hardwood lignin. These findings mark a significant advancement in converting renewable biomass into high-value methoxylated cycloalkanes via electrocatalysis, underscoring the pharmaceutical relevance and strategic importance of this research for bioproduct synthesis and green chemistry. This solid method provides a promising basis for future sustainable biomass valorization in research and industry [127].

This study investigates the integration of pyrolysis and electrocatalysis for lignin valorization in biorefineries, focusing on selective hydrogenation/hydrogenolysis of lignin-derived compounds to create fuels and chemicals. The phenolic-rich bio-oil from lignin fast pyrolysis was evaluated as an intermediate feedstock for ECH, improving aromatic ring saturation, deoxygenation, and energy content. Hydrogenation reactions of phenol, guaiacol, syringol, and various alkylphenols utilized Ru/ACC, aiming at C–O and C–C bond hydrogenolysis. Guaiacol derivatives formed alkyl-substituted 2-methoxycyclohexanols with limited demethoxylation. Increased alkyl chain lengths slowed reactions and altered product selectivity toward methoxylated cycloalkanes. The conversion of cresol derivatives was enhanced by methyl proximity to hydroxyl groups, with neighboring –OH and –OCH₃ notably improving conversion and selectivity. Trace alkylcyclohexanes resulted from C–OH bond cleavage in high-molecular-weight compounds, increasing with longer alkyl chains. Higher substrate concentrations boosted Faradaic efficiency to a plateau, emphasizing the importance of feedstock loading and catalyst interactions in ECH systems. This research outlines a lignin valorization method involving biomass fractionation, pyrolysis, and electrocatalytic upgrading, demonstrating that the structure of phenolic compounds influences selectivity, kinetics, and electrochemical efficiency, and suggesting that combining lignin pretreatment with electrochemical stabilization could enhance biorefinery strategies for transforming lignin-based bio-oils into valuable products [129].

This study presents a novel method for the electrochemical conversion of levulinic acid (LA), a biomass-derived compound, into high-energy biofuel precursors. The authors

demonstrate that LA can be transformed into compounds like valeric acid (VA) and γ -valerolactone (gVL) using a low-cost lead (Pb) electrode in a single-polymer electrolyte membrane electrocatalytic flow reactor. Achieving over 90% VA yield and more than 86% Faradaic efficiency, the process has an energy storage efficiency (ESE) of 70.8%, allowing efficient conversion of renewable electricity to chemical energy at a low energy requirement of $1.5 \text{ kWh} \cdot \text{L}^{-1}$. Product selectivity is affected by applied potential and electrolyte pH; lower overpotentials and neutral pH enhance gVL formation, while higher potentials and acidic conditions favor VA. This reactor surpasses conventional half-cell systems in conversion and efficiency due to enhanced mass transfer and optimized parameters. LA reduction involves a four-electron process, contrasting with other biomass derivatives which use two-electron transfers, highlighting how molecular structure impacts reduction kinetics. This eco-friendly method supports renewable electricity storage in chemical fuel form and offers a foundation for advanced catalysts and reactor engineering for producing hydrogen-rich biofuels [131].

A systematic investigation was conducted on the electrochemical conversion of levulinic acid (LA) in aqueous media at ambient temperature and pressure, using both oxidative and reductive pathways. In line with green chemistry principles, primary products like valeric acid (VA), γ -valerolactone (gVL), and 2,7-octanedione were synthesized, leading to secondary compounds relevant for fuel production, including n-octane and 1-butanol. Product selectivity was significantly influenced by factors such as electrode type, electrolyte composition, pH, and substrate concentration; for example, gVL selectivity reached 70% with iron electrodes in alkaline conditions. Notably, this study documents the electrochemical generation of 1-butanol from LA via a two-step pathway. Certain products' insolubility enabled direct phase separation and electrolyte recycling, showcasing low energy requirements and room temperature operation, indicating environmental and economic feasibility. This method demonstrates the potential for renewable electricity storage in liquid organic compounds, aligning with sustainability objectives [132].

Another study explored the electrocatalytic hydrogenation (ECH) of furfural to furfuryl alcohol. The cathode material choice significantly influenced selectivity and efficiency in an H-type cell using platinum foil as the anode. Platinum (Pt), nickel (Ni), copper (Cu), and lead (Pb) were tested, with Pt showing superior selectivity. A platinum-supported activated carbon fiber (Pt/ACF) electrocatalyst was then created through impregnation and electrodeposition, with the impregnated Pt/ACF revealing enhanced activity due to its larger surface area. Notably, the 3% Pt-loaded ACF outperformed the 5% version in current and product efficiency. The electrolyte type proved crucial, with 0.1 M HCl resulting in the highest production of furfuryl alcohol. Key variables included furfural concentration and applied potential, with an optimal potential of -0.5 V found for the 3% Pt/ACF catalyst. These findings highlight the effectiveness of Pt/ACF for selective ECH of furfural, supporting an energy-efficient approach to sustainable furfuryl alcohol production [134].

This study investigates the electrocatalytic hydrogenation and hydrogenolysis (ECH) of furfural (FF) on a copper (Cu) electrode in an H-type batch reactor at room temperature, highlighting how reaction selectivity for biofuels and fine chemicals varies with operating conditions. A correlation between product distribution and parameters like current density, electrolyte acidity, co-solvent ratio, and electron transfer was found. The electrolyte pH was critical: furfuryl alcohol (FA) was predominant under mildly acidic conditions (0.2 M NH_4Cl), while FA and 2-methylfuran (MF) appeared in strongly acidic media (0.1–0.5 M H_2SO_4). Increasing current density improved the product yield without notably affecting the distribution in four-electron transfer reactions. The overall mass balance stayed below 70% due to side reactions and charge transfer at the catalyst surface. In sulfuric acid, FA,

MF, and FF likely turned into undetectable products. In NH_4Cl , higher current densities minimized side reaction duration, improving yield and selectivity. Although acetonitrile addition enhanced yield at times, no consistent trend was noted. This study emphasizes optimizing electrolyte type, pH, reaction time, and separation strategies for selective FA and MF production from FF via ECH, as efficient MF extraction could enhance yield while reducing byproducts [135].

A significant study presents lanthanum (La^{3+})-doped nano- TiO_2 film electrodes for the electrocatalytic reduction of furfural to furfuryl alcohol, showcasing a novel application. The electrodes, made via sol-gel methods, were characterized using XRD, FE-SEM, XPS, and cyclic voltammetry (CV), showing smaller particle sizes and improved morphology. XRD and XPS confirmed La^{3+} integration into the TiO_2 lattice. The La-doped electrodes displayed higher electrocatalytic activity than undoped TiO_2 , achieving 88.6% yield and 85.7% Faradaic efficiency for furfuryl alcohol in N,N-dimethylformamide (DMF), as verified by NMR. This positions La-doped TiO_2 films as effective platforms for the electroreduction of furfural, with potential in broader electrocatalysis [136]. This research establishes a correlation between the fundamental catalytic properties of electrocatalytic hydrogenation (ECH) and material descriptors, crucial for integrating recycled carbon sources with renewable energy. Previous studies generally lacked clear links between substrate or catalyst characteristics. This work evaluated various substrates (aldehydes, ketones, carboxylic acids, and phenolic compounds) with different metal electrodes (Pd, Rh, Ru, Cu, Ni, Zn, and Co), connecting reaction rates to substrate properties. A key discovery was that aldehyde reduction rates showed a volcano-type relation to metal-substrate binding energies, consistent with the Sabatier principle; optimal activity occurs with neither weak nor strong adsorption. Additionally, hydrogen evolution reaction (HER) rates correlated with hydrogen-metal binding strengths, indicating shared catalytic principles between ECH and HER. Correlating DFT-calculated binding energies with experimental rates clarified kinetic behavior and supported the predictive modeling of metal reactivity. Pd showed the highest activity for aromatic aldehyde reduction due to optimal binding affinity, providing a predictive framework for designing monometallic and bimetallic systems [137].

A recent study focused on developing a stable, high-performance electrocatalyst for the electrochemical hydrogenation (ECH) of guaiacol and related lignin-derived monomers to improve bio-oil upgrading. The PtNiB/CMK-3 nanocomposite features boron-doped PtNi alloy nanoparticles on mesoporous CMK-3 carbon, which altered the electronic structure of the alloy and enhanced reactivity. Experimental and theoretical analyses revealed that B doping increased Faradaic efficiency to 86.2%, a 13.7-fold improvement over the undoped catalyst. The CMK-3 support, with its high surface area and ordered pores, improved active site exposure and mass transfer, leading to better catalyst stability and yield. The even distribution of PtNiB nanoparticles within CMK-3 enhanced utilization and scalability. Laboratory tests demonstrated efficient guaiacol conversion into valuable chemicals and KA oil, showcasing a promising electrocatalyst for sustainable fine chemical production [139].

Recent research has shown that the electrochemical hydrogenation of biomass-derived compounds is significantly influenced by the surface coverage and reactivity of adsorbed hydrogen species (H^*). One study indicated that the interaction between electrochemical hydrogenation (ECH) and the hydrogen evolution reaction (HER) is primarily determined by how both H^* and the substrate adhere to the electrocatalyst surface. In hydrogen-scarce environments, the limited presence of H^* can lead to undesirable side reactions, such as dimerization. Conversely, hydrogen-abundant conditions encourage HER, which consequently decreases the Faradaic efficiency (FE) of ECH. Thus, maintaining an optimal surface coverage of H^* is crucial for enhancing both the FE and selectivity of ECH processes [43,117]. The review further elaborates on the mechanisms underlying ECH by comparing two main

pathways: the Eley–Rideal (E–R) mechanism, where surface-adsorbed organic molecules react with free protons or hydrogen from water, and the Langmuir–Hinshelwood (L–H) mechanism, in which both hydrogen atoms (H^*) and the organic substrates are co-adsorbed on the electrode surface before hydrogenation. These mechanisms are directly related to the hydrogen adsorption energy (ΔG_H) and are significantly affected by the choice of electrode material, as demonstrated by density functional theory (DFT) calculations and in situ electrochemical analyses. For instance, catalysts such as copper (Cu), silver (Ag), and palladium (Pd) can adjust the binding energy of H^* , thereby modifying the balance between the rates of hydrogenation and HER, which provides a potential approach to enhancing catalytic effectiveness [140–142].

The authors have shown that optimizing electrode design, particularly in flow-type and H-cell configurations, can enhance Faradaic efficiency (FE) by reducing the accumulation of H_2 bubbles and improving control over the local hydrogen (H^*) environment. Operando electrochemical impedance spectroscopy (EIS) and electron paramagnetic resonance (EPR) studies have highlighted the simultaneous occurrence of Volmer and Heyrovsky steps during the hydrogen evolution reaction (HER) and electrochemical hydrogenation (ECH). Additionally, the surface states of catalysts, such as Cu^+/Cu^0 or single Ru atoms, significantly affect the generation, transfer, and use of H^* . These insights indicate that refining the adsorption and reaction characteristics of H^* is essential for elucidating the mechanistic pathways of ECH and for the strategic design of electrocatalysts aimed at enhancing selectivity, efficiency, and sustainability. Research has provided strong evidence that adjusting the density and binding strength of hydrogen on non-precious metal electrodes, including copper (Cu), silver (Ag), and gold (Au), facilitates high catalytic performance under mild electrochemical conditions. Comprehensive experimental and theoretical studies indicated that achieving a hydrogen coverage of 0.75 monolayers on Cu electrodes promotes the selective hydrogenation of acetophenone to 1-phenylethanol, with Faradaic efficiencies peaking at 91%, while concurrently inhibiting the competing HER. This observation aligns with Sabatier’s principle and aligns with previous studies revealing that moderate hydrogen adsorption free energies (ΔG_H) are advantageous for selective ECH processes [142–145]. The researchers also identified a robust inverse relationship between HER activity and H^* depletion during the ECH process, consistent with earlier kinetic analysis on platinum (Pt) and Cu electrodes [146–148]. Furthermore, mechanistic perspectives derived from density functional theory (DFT) calculations suggest that water molecules act as proton carriers, facilitating H^* transfer, and that the adsorption of organic substrates can influence H^* availability by competing for active surface sites [149–151]. Overall, these findings highlight hydrogen surface coverage not merely as a mechanistic aspect but as a vital design factor for the development of advanced ECH electrocatalysts. By integrating theoretical approaches with experimental findings and situating these results within the broader context of electrocatalysis, this research establishes a foundation for the rational progression of catalysts aimed at achieving sustainable and selective electrochemical valorization of organic compounds derived from biomass.

Table 2 below summarizes the key differences between electrochemical hydrogenation (ECH) and thermochemical hydrogenation (TCH) processes. It highlights various aspects such as energy efficiency, selectivity, reaction conditions, catalyst performance, surface behavior, and industrial applicability. This comparative overview serves to illustrate the advantages and limitations of each method, providing insights into their potential for sustainable and scalable hydrogenation of biomass-derived compounds.

Table 2. Comparative evaluation of electrochemical and thermochemical hydrogenation processes.

	Electrochemical Hydrogenation (ECH)	Thermochemical Hydrogenation (TCH)
Energy Efficiency	It efficiently employs electrical energy and requires only a small amount of thermal energy. This system can work with renewable energy sources, and energy recovery is possible through built-in half-reactions. For example, producing two outputs in a single cell has shown to result in energy savings of up to 50%.	The process demands the production of external hydrogen (H ₂) and heating, with the generation and compression of hydrogen leading to extra energy expenses. There is a restricted use of waste heat, and co-reactions are typically not taken advantage of, potentially causing a reduction in overall energy efficiency.
Selectivity	Low temperatures and adjustable potentials facilitate achieving high selectivity for the target product, simultaneously reducing side reactions. For instance, under optimized alloy composition and reaction conditions, a single-product selectivity exceeding 95% can be attained. Additionally, by adjusting the potential and the reaction environment, it is possible to select intermediate products by stopping the reaction at a predetermined stage.	When there is an adequate supply of hydrogen (H ₂), thermochemical hydrogenation typically facilitates a targeted transformation. Nonetheless, higher temperatures can heighten the likelihood of side reactions. In cases where the catalyst possesses acidic sites, unwanted processes like deoxygenation might take place. It is challenging to accumulate intermediate products unless the procedure is deliberately halted, given that the reaction naturally progresses towards the end product from a thermodynamic perspective.
Reaction Conditions	Functions effectively under gentle conditions, specifically at atmospheric pressure and within the temperature range of 20–60 °C. This method can be readily utilized in aqueous environments or in slightly organic-modified settings, as hydrogen is generated on-site from water, eliminating the need for an external gas supply. The procedure is fundamentally safe and does not necessitate intricate pressurized apparatus.	The process may necessitate the use of high-pressure hydrogen gas, usually ranging from several bar to twenty bar, depending on the specific reaction conditions. Elevated temperatures, typically exceeding 100 °C, are frequently utilized. In certain cases, organic solvents or biphasic systems might be required, with the solubility of hydrogen being a potential constraint. The complexity and expense of the operation are increased due to the need for specialized pressurized equipment and heating systems.
Catalyst Activity and Reusability	The low operational temperatures significantly reduce the risks of catalyst sintering and coke accumulation, leading to prolonged catalyst activity. Catalysts are typically very stable; for example, the Pt/SSB catalyst showcased a conversion rate of 94% after undergoing three cycles. Regeneration of the catalyst surface can be facilitated by re-hydrogenating the poisoning agents, which is accomplished by modifying the electrode potential. When the catalyst is applied to the electrode, it is straightforward to keep it contained within the reactor; alternatively, if the catalyst is in suspension, it can be easily filtered and reused.	Metal particles can sinter under thermal conditions, reducing the active surface area for reactions. Elevated temperatures can create byproducts like coke and polymers, contaminating the catalyst surface. Periodic regeneration methods, such as hydrogen treatment or oxidative cleaning, may be needed. Extracting solid catalysts from pressurized reactors is challenging since they are typically in fixed-bed setups and replaced upon deactivation. Nonetheless, many commercial catalysts can work effectively for hundreds of hours, with their stability influenced by feedstock purity and operating temperature.

Table 2. Cont.

	Electrochemical Hydrogenation (ECH)	Thermochemical Hydrogenation (TCH)
Surface Poisoning and Adsorption	The buildup of specific intermediate species on a catalyst’s surface can result in catalyst poisoning, primarily due to insufficient hydrogen (H) coverage. This issue can be alleviated by either modifying the applied potential or increasing the proton flux, as indicated in the literature. For example, during electrochemical hydrogenation (ECH) processes, raising the current or potential enhances the rehydrogenation of phenolic species that are adsorbed, thus helping to restore the electrode surface. Additionally, organic surface modifiers like carboxylates can be intentionally adsorbed onto the electrode to influence the reaction pathway. Furthermore, the adsorption strength of hydrogen on the metal surface can be adjusted through alloying, which enables the optimization of the relationship between the hydrogen evolution reaction (HER) and electrochemical hydrogenation (ECH).	At high temperatures, hydrocarbon derivatives may polymerize on catalysts, leading to deactivation. To address this, elevated hydrogen (H₂) partial pressures are maintained, which helps to prevent poisoning and allows regeneration by hydrogenating accumulated species. In thermal catalytic hydrogenation (TCH), the active surface’s potential is not externally controlled; instead, the adsorption characteristics depend on the catalyst formulation. Metals like palladium (Pd) can form hydride phases under TCH, affecting reaction kinetics, though this is less significant than in electrochemical catalytic hydrogenation (ECH). Catalyst poisoning regeneration often involves increasing H₂ pressure or replacing the deactivated catalyst.
Industrial Applicability and Scalability	Modular and Scalable Design: Electrochemical cells can be arranged in series or parallel configurations to meet specific capacity requirements. They utilize similar infrastructure to current technologies, including water electrolysis and electroplating. Safety Benefits: Since high-pressure hydrogen is not stored, the likelihood of explosion is significantly reduced. When powered by renewable energy sources, this process is completely sustainable. The operation occurs at low temperature and pressure, leading to decreased equipment costs. Moreover, it is well suited for decentralized applications, such as on-site wastewater treatment. An example of this is the effective electrochemical hydrogenation (ECH) of phenol and guaiacol in continuous flow and membrane reactor systems, which has established a framework for larger-scale implementation.	Traditional and Commonly Employed in Industry: Fixed-bed reactors, batch autoclaves, and continuous reactors are available for large-scale applications. These systems necessitate hydrogen supply and storage, making them more appropriate for centralized, large facilities. Processing feedstocks with low hydrogen concentrations is not efficient; for instance, hydrogenating phenol in wastewater under high pressure is not feasible. Most current processes rely on hydrogen derived from fossil fuels, resulting in a substantial carbon footprint. Although the implementation of green hydrogen is an option, it tends to be expensive. At present, economies of scale favor large-volume production, but this situation may not be beneficial for small-scale or decentralized operations.

Electrocatalytic hydrogenation (ECH) generally operates within a temperature range of 25–60 °C and is conducted under atmospheric pressure [113–115]. Such moderate operational conditions negate the necessity for heating and pressurization equipment, making ECH particularly advantageous for systems where high energy density is not a critical requirement. In stark contrast, thermochemical hydrogenation (TCH) necessitates elevated temperature conditions (100–250 °C) and typically requires high hydrogen gas pressures, often ranging between 5 and 30 bar [115], thereby demanding the use of more intricate and expensive equipment, along with increased energy input and greater safety concerns.

As a result, ECH provides significant benefits in terms of energy efficiency, lower operational costs, and improved safety during operation. In ECH systems, the reaction pathway is amenable to precise control via adjustments in applied potential, electrolyte composition, and pH levels [110,117,125,126]. This degree of control allows for the meticulous management of reactions. For instance, the selective hydrogenation of aromatic compounds can be fine-tuned to either achieve partial saturation of the aromatic ring (as exemplified by the transformation of phenol to cyclohexanone) or attain full saturation (for instance, the production of cyclohexanol) simply by varying the applied potential [118]. Such precise control diminishes the likelihood of undesired byproduct formation while facilitating the selective accumulation of reaction intermediates (such as cyclohexanone), a feat that is typically more challenging in TCH processes, where temperature and catalyst acidity predominantly dictate selectivity. Another notable advantage of ECH lies in its capacity to generate hydrogen in situ through water electrolysis, thereby negating the requirement for external H₂ gas [111,119].

This characteristic effectively addresses challenges linked to hydrogen transportation, storage, and compression activities that are often hazardous and require a significant energy commitment. Additionally, ECH can be directly powered by renewable energy sources, resulting in a carbon-neutral process [121]. Conversely, TCH predominantly depends on hydrogen derived from fossil sources, consequently amplifying its overall carbon footprint. This positions ECH as a more environmentally sustainable conversion pathway. A primary challenge associated with ECH is the competition posed by the hydrogen evolution reaction (HER), in which surface-adsorbed hydrogen atoms (H*) tend to recombine to generate H₂ gas [101]. However, this challenge can be partially alleviated by the utilization of tailored alloy catalysts (such as PtRh, PtSn, or PtRu) and advanced support materials (for instance, high-surface-area carbon or nitrogen-doped biochar) [117,118].

By optimizing the system to promote the reaction of H* with organic molecules over HER, Faradaic efficiencies as high as 86% can be realized, significantly enhancing the efficiency of electrochemical transformations. Furthermore, the electrode surfaces utilized in ECH systems can be chemically and functionally modified to augment both reactivity and selectivity. For example, the application of surface modifications, such as carboxylic acids, has been shown to improve the adsorption and conversion efficiency of molecules like phenol [112]. Such deliberate surface chemistry fosters the desired reactions while inhibiting undesired pathways, an opportunity that is often less viable in TCH due to the destabilization of surface modifications at elevated temperatures. Moreover, ECH provides the unique advantage of reactivating and eliminating adsorbed inhibitory species from the electrode surface through the application of controlled potential [115].

This feature is particularly beneficial in scenarios where partially hydrogenated or oxidized organic compounds may obstruct active sites. In contrast, TCH often experiences catalyst deactivation resulting from carbon accumulation, coking, or metal sintering, necessitating the offline regeneration of reactor beds. Consequently, ECH systems tend to exhibit longer operational lifespans and reduced maintenance demands. Finally, ECH systems demonstrate high scalability and can be effectively implemented using series-parallel cell configurations. The deployment of flow-type cells, stirred suspension electrochemical reactors (SSERs), and three-dimensional structured electrodes allows for a smooth transition from laboratory settings to pilot scales. This modularity fosters decentralized production approaches and mitigates reliance on centralized facilities. In contrast, TCH processes typically necessitate the use of large, fixed-bed reactors, which imposes constraints on spatial flexibility and portability.

5. Conclusions

This study presents a comprehensive examination of the environmental and technological advantages of electrochemical hydrogenation (ECH) processes and thermochemical conversion in the conversion of bio-oils derived from biomass sources. The complex characteristics of lignocellulosic biomass offer a promising feedstock for bio-oil production, yet challenges such as elevated oxygen content, acidity, and chemical instability necessitate sophisticated upgrading methodologies. ECH technologies are particularly noteworthy due to their operation under mild temperatures and pressures, which result in high product selectivity and improved energy efficiency. In contrast, traditional thermal hydrogenation approaches are often associated with substantial energy demands and potential carbon emissions, thus raising sustainability concerns regarding fossil-based methodologies. ECH technology's capacity to utilize renewable electricity as a hydrogen source allows for a considerable reduction in greenhouse gas emissions during conversion processes.

Consequently, ECH serves a crucial role in mitigating carbon footprints and tackling climate change challenges. Furthermore, the advantages of ECH processes extend to high reaction selectivity, minimal byproduct production, precise control over reaction conditions, and modular scalability, thus establishing them as an appealing alternative in sustainable chemistry applications across both academic and industrial domains. In particular, advancements in PEM (Proton-exchange Membrane) technology, as well as membrane and catalyst innovations employing graphene-based materials with high surface areas, exhibit promising improvements in reaction kinetics and selectivity. Nevertheless, several technological and economic hurdles must be addressed to facilitate the wider industrial implementation of ECH systems. Critical factors influencing system performance include catalyst deactivation, reactor design optimization, hydrogen transport inefficiencies, and surface coating imbalances.

Future research endeavors should prioritize enhancing catalyst longevity, refining adsorption–desorption processes, and integrating energy-recovery methodologies. In summary, electrochemical hydrogenation emerges as an environmentally responsible, economically advantageous, and technologically viable solution for the advanced conversion of renewable carbon resources such as biomass. Given its potential for low carbon emissions, energy efficiency, and environmental safety, ECH processes are poised to play a fundamental role in future green chemistry and energy technologies. To promote broader sectorial adoption, it is imperative to encourage public–private collaborations, undertake pilot-scale demonstrations, and support interdisciplinary research initiatives that will further explore promising PEM and membrane technologies in conjunction with ECH systems.

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